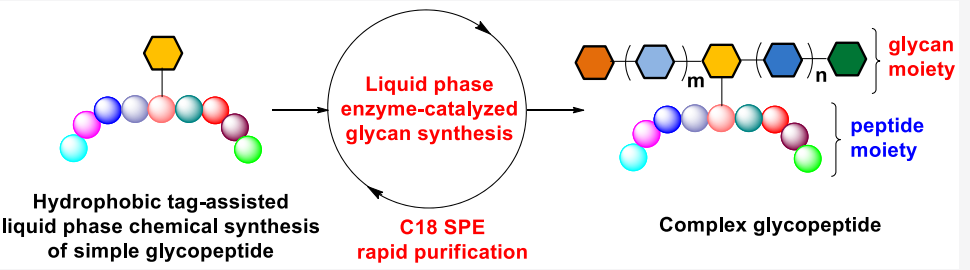
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Integrated Chemoenzymatic Approach to Streamline the Assembly of Complex Glycopeptides in the Liquid Phase

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Cite This: J. Am. Chem. Soc. 2022, 144, 9057−9065
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ABSTRACT: Glycosylation of proteins is a complicated post-translational modification. Despite the significant progress in glycoproteomics, accurate functions of glycoproteins are still ambiguous owing to the difficulty in obtaining homogeneous glycopeptides or glycoproteins. Here, we describe a streamlined chemoenzymatic method to prepare complex glycopeptides by integrating hydrophobic tag-supported chemical synthesis and enzymatic glycosylations. The hydrophobic tag is utilized to facilitate peptide chain elongation in the liquid phase and expeditious product separation. After removal of the tag, a series of glycans are installed on the peptides via efficient glycosyltransferase-catalyzed reactions. The general applicability and robustness of this approach are exemplified by efficient preparation of 16 well-defined SARS-CoV-2 *O*-glycopeptides, 4 complex MUC1 glycopeptides, and a 31-mer glycosylated glucagon-like peptide-1. Our developed approach will open up a new range of easy access to various complex glycopeptides of biological importance.

INTRODUCTION

Glycosylation, a common but complicated post-translational modification, is implicated in diverse physiological and pathological activities including cell adhesion and signaling, bacterial and viral infections, immune evasions, and cancer metastasis.[1−3] Majority of the human proteins are glycosylated,[4] and biological functions of glycopeptides are realized by the collaborative effects of both the glycan structure and peptide sequence.[5,6] Although elegant synthesis methods[7−10] and advanced analysis technologies for glycomics and proteomics[11,12] have been established, the precise functional studies of glycopeptides are still impeded by lack of robust and cost-effective synthetic methods to prepare a wide range of structurally well-defined glycopeptides. These compounds can serve as probes for elucidating their biosynthesis mechanisms and standards for identifying accurate structures of glycopeptides in complicated biological mixtures and offer opportunities to conduct high-throughput microarray analysis for binding studies,[13−15] which could contribute to developing therapeutic agents and vaccines.[16−18]

The availability of peptides widely relies on solid-phase peptide synthesis (SPPS). In the process of SPPS, chemical elongation of the peptide chain was achieved by successive couplings and deprotections on a resin, and each separation was performed by rapid filtration.[19] In a manner similar to SPPS, preparation of glycopeptides bearing simple mono- or disaccharide is readily accomplished using glycosylated amino acids to elongate the peptide chain.[20−22] Nevertheless, the synthesis of complex glycopeptides carrying large-sized and complicated glycan is still challenging[23] due to the need for chemical preparation of the glycosylated amino acid moiety, which involves difficulties in stereoselective construction of sialic acid linkages, tedious protection−deprotection manipulations, and time-consuming chromatographic purification procedures.[24,25] In addition, either the β-eliminations or the cleavages of acid-sensitive sialoside and fucoside may occur during the glycopeptide deprotection process.[26−29] Alternatively, the glycosyltransferase-mediated reaction usually enables regio- and stereoselective glycosylation,[30] providing opportunities for extension of the simple sugars, which have been

Received: February 16, 2022
Published: May 11, 2022



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https://doi.org/10.1021/jacs.2c01819
J. Am. Chem. Soc. 2022, 144, 9057−9065

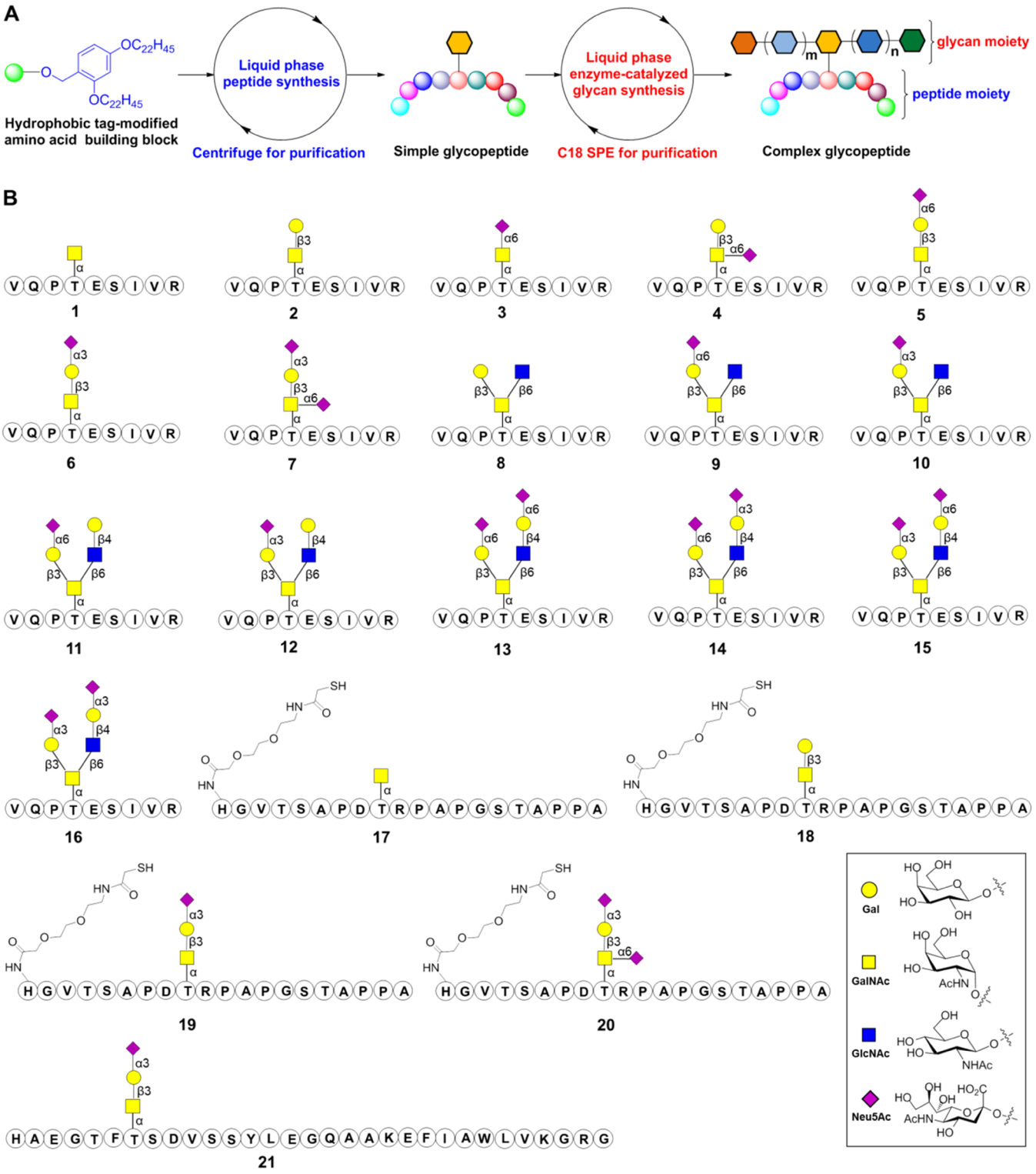
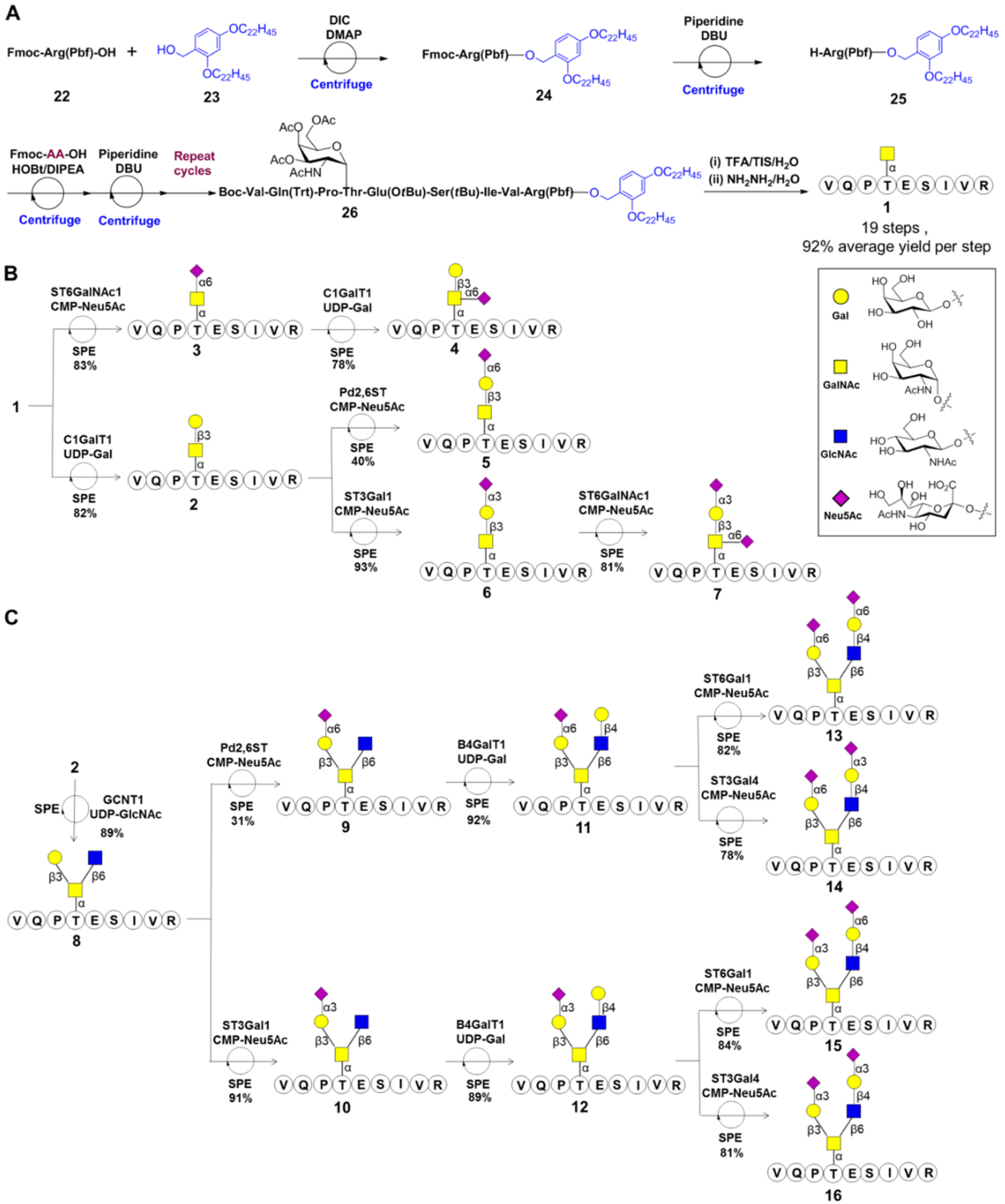


Figure 1. Integrated chemoenzymatic approach to efficiently synthesize complex glycopeptides in the liquid phase. (A) Hydrophobic tag-assisted peptide chain growing and glycosyltransferase-catalyzed glycan extension to afford complex glycopeptides (the colored spheres represent different amino acids, and the colored hexagons represent different monosaccharides). (B) Synthetic glycopeptide library including 16 SARS-CoV-2 *O*-glycopeptides, four 20-mer MUC1 glycopeptides with a mercapto-functionalized spacer, and a 31-mer glycosylated glucagon-like peptide-1.

grafted on the peptide backbone to generate various complex glycan units. The Wong group has pioneered the application of an aminopropyl-functionalized silica resin, which is suited for both organic- and aqueous-phase reactions, to achieve chemoenzymatic preparation of glycopeptides with a complex tetrasaccharide sialyl Lewis^{X}.[31] Recently, a semiautomated platform using a peptide synthesizer was developed to chemoenzymatically synthesize *O*-glycopeptides based on this organic−aqueous-compatible silica resin.[32] However, the high efficiency and scope of enzyme-catalyzed reactions by this solid-supported procedure of immobilizing glycosyl acceptors on the resin for sugar chain elongation are limited because

https://doi.org/10.1021/jacs.2c01819
J. Am. Chem. Soc. 2022, 144, 9057−9065

Scheme 1. Divergent Chemoenzymatic Synthesis of Diverse SARS-CoV-2 *O*-Glycopeptides[a]



[a]**C1GalT1:** GalNAc-β1,3-galactosyltransferase **1**; **ST6GalNAc1:** GalNAc-α2,6-sialyltransferase **1**; **ST3Gal1:** Gal-α2,3-sialyltransferase 1; **GCNT1:** β1,6-GlcNAc-transferase **1**; **B4GalT1:** β1,4-galactosyltransferase **1**; **ST6Gal1:** Gal-α2,6-sialyltransferase **1**, **ST3Gal4:** Gal-α2,3-sialyltransferase **4**; **Pd2,6ST:** *Photobacterium damselae* α2,6-sialyltransferase; and **SPE:** solid-phase extraction.

glycosyltransferases are relatively large and cannot easily migrate into the pores of the resin to catalyze glycan synthesis, resulting in substantially hydrolyzing the utilized sugar nucleotide donors.[33] To address this issue, chemoenzymatic

https://doi.org/10.1021/jacs.2c01819
J. Am. Chem. Soc. 2022, 144, 9057−9065

approaches have been introduced by combining solid-phase peptide synthesis on the resins and liquid-phase enzyme-mediated glycan elongation on water-soluble polymers to generate complex glycopeptides.[34,35] Compared with *O*-glycopeptide synthesis, the preparation of complex *N*-glycopeptide often starts with the synthesis of GlcNAc-peptide as an acceptor by SPPS, followed by the endoglycosidase-mediated ligation of a large-sized *N*-glycan oxazoline donor that requires a challenging synthesis.[36,37] Despite that these excellent improvements for the solid/polymer-supported synthesis have been implemented to prepare various glycopeptides, these methods are still restricted by the low scale, the difficulty in monitoring the progress of reactions, and the usage of an excessive amount of expensive reagents, especially the glycosylated amino acids.

Liquid-phase peptide synthesis (LPPS) provides a potential option for SPPS.[38] LPPS can break scale limitation and monitor the reaction in real time. Unlike SPPS, LPPS does not require a substantial excess of reagents to push the coupling reaction to completion, which greatly improves the reaction efficiency in a relatively economical manner. Additionally, the separation process in LPPS can be accomplished by extraction or precipitation based on tag technologies[39−43] and diafiltration technologies.[44,45] Recently, a novel liquid peptide synthesis approach was introduced by Livingston and Albericio via one-pot nanostar sieving.[46] Moreover, several automated solution-phase synthesis platforms such as fluorous tag-assisted synthesis of glycans,[47] sulfonate tag-supported enzymatic assembly of complex oligosaccharides,[33] and fast-flow synthesis of deoxy-sugars[48] and oligonucleotides[49] have been developed to enable easy access to various biomolecules. Given these impressive advancements in the liquid-phase assembly of peptides, oligosaccharides, and oligonucleotides, we reported here a chemoenzymatic approach to streamline the preparation of diverse and complex glycopeptides in the liquid phase (Figure 1). The strategic point of this method is that efficient chemical elongation of the peptide chain bearing a simple sugar is performed in the liquid phase and the separation is accomplished by a hydrophobic tag-assisted expeditious centrifugation, during which only 1.2 equiv of simple glycosylated amino acid building blocks and coupling reactants is required, and the progress of the reactions are readily monitored. More importantly, after removing the tag, the enzymatic extension of the resulting glycosyl acceptor peptide using a panel of robust glycosyltransferases is efficiently carried out in the aqueous phase, and the separation is accomplished through a rapid solid-phase extraction (SPE) using a C18 cartridge. The general applicability and robustness of this approach are exemplified by the streamlined synthesis of various complex glycopeptides including 16 SARS-CoV-2 *O*-glycopeptides, four 20-mer MUC1 glycopeptides with a mercapto-functionalized spacer, and a 31-mer glycosylated glucagon-like peptide-1.

RESULTS AND DISCUSSION

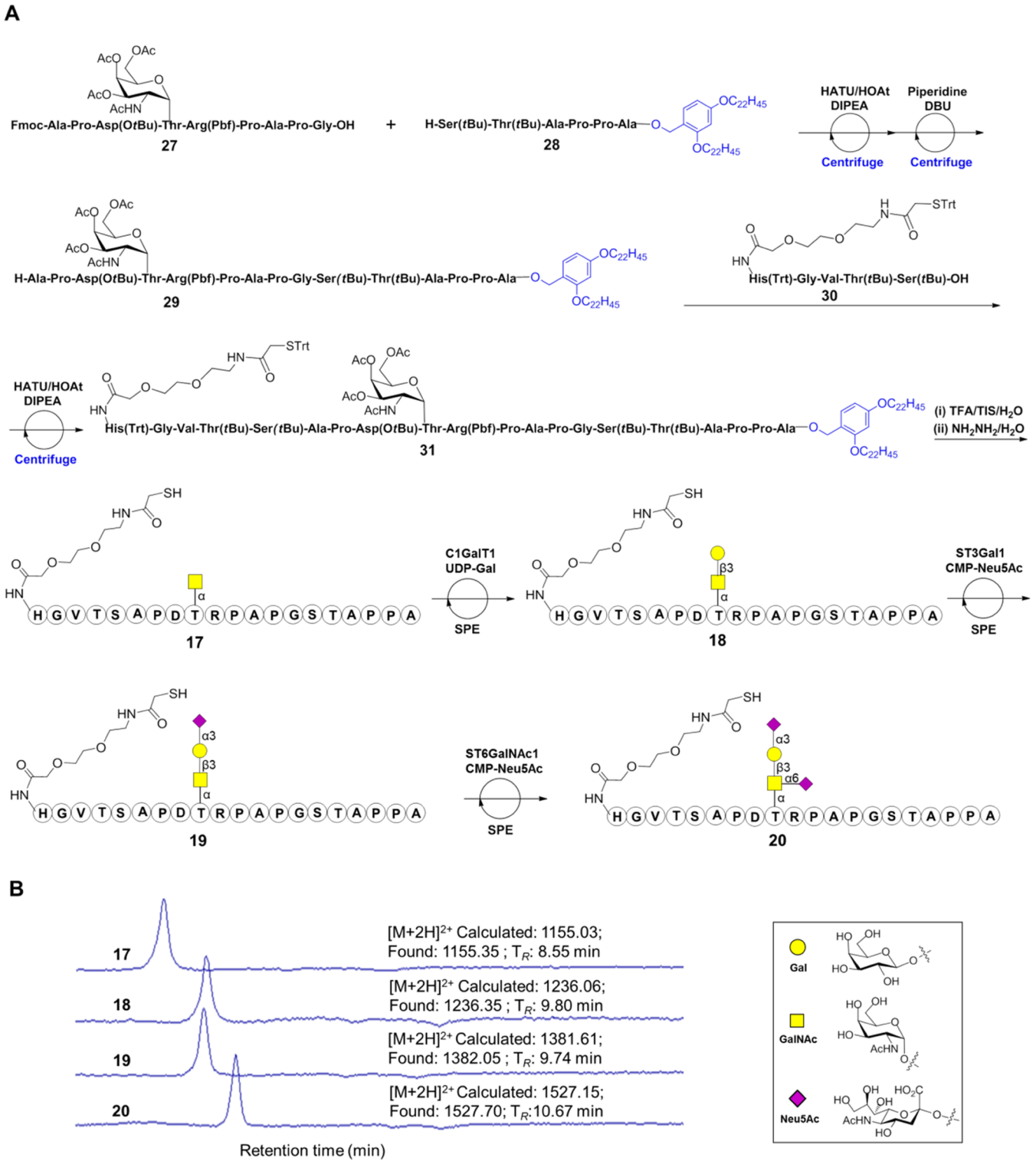
Divergent Chemoenzymatic Synthesis of Diverse SARS-CoV-2 *O*-Glycopeptides. To demonstrate the feasibility of our method for the linear chemical elongation of peptides and the divergent enzymatic extension of glycans to assemble complex glycopeptides in the liquid phase, we prepared a collection of SARS-CoV-2 *O*-glycopeptides located at the receptor binding domain (RBD). Recently, glycomics studies have revealed that *O*-glycans on site Thr323 are structurally diverse, which are mainly constituted by cores 1 and 2 *O*-GalNAc oligosaccharides with further mono- or disialylated modifications.[50] The heterogeneity and structural complexity of glycans make it difficult to obtain homogeneous *O*-glycopeptides isolated from SARS-CoV-2 in sufficient quantities with high purity, which severely restricts the related function studies of *O*-glycosylations. Alternatively, our synthetic approach provides access to well-defined *O*-glycopeptides. The synthesis of *O*-glycopeptides **1**−**16** started from the protected Arg residue **22**, which was coupled with the hydrophobic tag **23** using *N*,*N*′-diisopropylcarbodiimide (DIC) and catalytic 4-dimethylaminopyridine (DMAP). The tag-modified product **24** was quantitatively obtained as a precipitate by dilution using a polar solvent (CH_3CN) and separated by quick centrifugation for the removal of excess amino acids and coupling reagents. Using general Fmoc chemistry and this hydrophobic tag-assisted liquid synthesis, compound **25** (0.4 mmol) underwent successive couplings and deprotections to afford tag-modified peptide **26**, during which the purification was performed by the similar centrifugation procedure described above due to the well-formed precipitate in each step. The acid-labile tag and all acid-sensitive side chain-protecting groups such as *tert*-butoxycarbonyl (Boc), trityl (Trt), *tert*-butyl (*t*Bu), and 2,2,4,6,7-pentamethyldihydrobenzofuran-5-sulfonyl (Pbf) were easily cleaved in a mixture solution of TFA/TIS/H_2O (38:1:1, Figure S2). Subsequently, three acetyl (Ac) groups in sugar residue were removed using 10% hydrazine in water to provide a crude product, followed by further purification through a C18 column to afford glycopeptide **1** (105 mg, 85 μmol) with 21% overall yield for 19 steps (Scheme 1A).

Next, we turned our interests to divergent enzymatic synthesis of various cores 1 and 2 *O*-GalNAc glycopeptides **2**−**16** containing monosialylated glycopeptides **4**−**6** and disialylated glycopeptides **13**−**16** (Scheme 1). Glycosyltransferases including β1,3-galactosyltransferase **1** (C1GalT1), α-2,6-sialyltransferases (ST6GalNAc1, ST6Gal1, and Pd2,6ST), α2,3-sialyltransferases (ST3Gal1 and ST3Gal4), β-1,6-GlcNAc-transferase **1** (GCNT1), and β1,4-galactosyltransferase **1** (B4GalT1) have been identified to be effective for synthesizing *O*-GalNAc oligosaccharides.[10,51−53] Therefore, it was expected that the GalNAc moiety of **1** could be extended by these glycosyltransferases to afford complex glycopeptides **2**−**16**.

Due to the enzyme-catalyzed reaction performed in the aqueous phase, we can readily monitor the reaction process with electrospray ionization−mass spectrometry (ESI-MS).[33,54] If the starting glycopeptide is detected, additional enzymes and sugar nucleotides are added to drive the reaction to completion. Then, the product was purified through a rapid solid-phase extraction (SPE) using a C18 cartridge to remove excess sugar nucleotides, enzymes, and buffer, followed by quantitative release of the glycopeptide. First, the sialylation of the GalNAc residue in glycopeptide **1** with ST6GalNAc1 and CMP-Neu5Ac provided α2,6-sialylated glycopeptide **3** in a yield of 83% (Scheme 1B). The GalNAc moiety of **3** was further galactosylated with C1GalT1 and UDP-Gal to afford glycopeptide **4**. Under the same reaction conditions, the galactosylation of **1** gave the desired glycopeptide **2** in a yield of 82%. The Gal residue of **2** was sialylated to give **5** (α2,6-linked Neu5Ac) with Pd2,6ST and CMP-Neu5Ac. It was discovered that Pd2,6ST-mediated α2,6-sialylation of the core 1 disaccharide moiety of **2** was slow. Consequently, more

https://doi.org/10.1021/jacs.2c01819
J. Am. Chem. Soc. 2022, 144, 9057−9065

Scheme 2. (A) Convergent Chemoenzymatic Synthesis of Complex 20-Mer MUC1 Glycopeptides 17−20 with a Mercapto-Functionalized Spacer and (B) LC-MS Analysis of Glycopeptides 17−20 after Each Enzymatic Transformation[a]

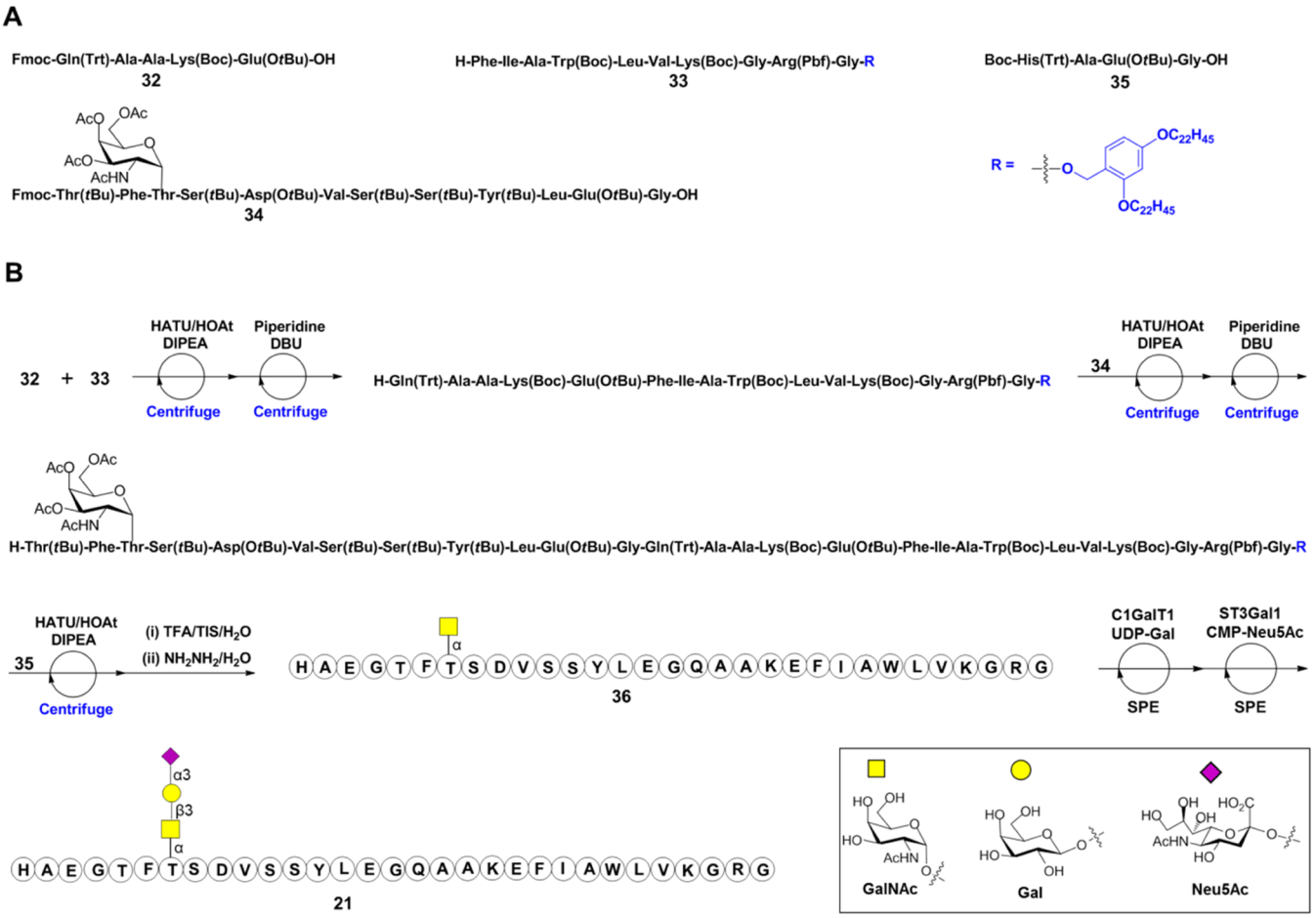


[a]**C1GalT1:** GalNAc-β1,3-galactosyltransferase **1**; **ST3Gal1:** Gal-α2,3-sialyltransferase **1**; **ST6GalNAc1:** GalNAc-α2,6-sialyltransferase **1**; and **SPE:** solid-phase extraction.

enzymes and CMP-Neu5Ac and longer incubation time are needed for pushing the transformation of the reaction. After the rapid C18 solid-phase extraction, the desired glycopeptide **5** was obtained with minor contamination of unreacted **2**, which was easily removed by semipreparative HPLC using a HILIC column (Table S2). In parallel, α2,3-sialylation of the Gal moiety of **2** using ST3Gal1 and CMP-Neu5Ac gave glycopeptide **6**. The internal GalNAc moiety of **6** was further sialylated using ST6GalNAc1 and CMP-Neu5Ac to give core 1 disialylated glycopeptide **7**.

https://doi.org/10.1021/jacs.2c01819
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Scheme 3. (A) Four Fragments 32−35 for Peptide Assembly by Hydrophobic Tag-Assisted Synthesis and (B) Convergent Chemoenzymatic Synthesis of a 31-Mer Glycosylated Glucagon-Like Peptide-1−21[a]



[a]**C1GalT1:** GalNAc-β1,3-galactosyltransferase **1**; **ST3Gal1:** Gal-α2,3-sialyltransferase **1**; and **SPE:** solid-phase extraction.

Subsequently, the selective installment of a GlcNAc residue to the internal GalNAc moiety of **2** afforded biantennary core 2 glycopeptide **8** using GCNT1 and UDP-GlcNAc (Scheme 1C). Treatment of **8** with Pd2,6ST and CMP-Neu5Ac provided glycopeptide **9** bearing α2,6-linked Neu5Ac.[53,55] Then, galactosylation of **9** using B4GalT1 and UDP-Gal afforded glycopeptide **11** in 92% yield. Sialylations of the terminal Gal residue of **11** using ST6Gal1- and ST3Gal4-mediated reactions provided the corresponding disialylated glycopeptides **13** and **14**, respectively. In parallel, treatment of **8** with ST3Gal1 and CMP-Neu5Ac gave α2,3-sialylated glycopeptide **10** in a yield of 91%. Next, using the same glycosylation sequence described above, elongation of the GlcNAc moiety of **10** with B4GalT1, ST6Gal1, ST3Gal4, and the corresponding sugar nucleotides successively afforded monosialylated glycopeptide **12** as the isomer of **11** and disialylated glycopeptides **15** and **16**. The resulting isomeric disialylated glycopeptides **13−16** were analyzed by HPLC on a HILIC column with ESI-MS for compound detection (Table S1). Gratifyingly, these four isomeric glycopeptides bearing different sialyl linkages can be separated on the HILIC column based on different retention times (Figure S20). This result demonstrated that these synthetic glycopeptides could serve as standards to facilitate the identification of exact glycopeptide structures in biological samples.

Convergent Chemoenzymatic Synthesis of Complex 20-Mer MUC1 Glycopeptides. To further demonstrate the applicability of the methodology, we turned our attention to the assembly of complex glycopeptides with a long peptide sequence. Given the difficulty and efficiency of the preparation of long glycopeptides, a convergent synthetic strategy is exploited by fragment couplings using the hydrophobic tag-supported synthesis, followed by glycan elongation using a panel of robust glycosyltransferases. The tumor-associated MUC1 glycopeptides, which are attractive targets for cancer vaccines and immunotherapeutics,[18] are chosen to be efficiently prepared. Our synthesized immunodominant glycopeptides are composed of 20 conserved amino acids (HGVTSAPDTRPAPGSTAPPA) that are glycosylated by various tumor-associated carbohydrate antigens including Tn, T, α2,3-sialyl T, and disialylated T.[56,57] To conjugate MUC1 glycopeptides with other biomolecules for developing cancer vaccines and therapeutic agents by thiol-maleimide chemistry, a functional linker containing a thiol moiety was introduced to glycopeptide *N*-terminus (Scheme 2). First, we rationally dissected the target peptide sequence into three fragments **27**, **28**, and **30**. After the preparation of hydrophobic tag-modified compounds (Figures S21−S23), **27** and **30** were smoothly obtained by selective removal of the tag in a mixture solution of TFA/TFE/DCM (1:10:100, Figures S22 and S23), which did not affect other acid-labile protecting groups. The coupling

https://doi.org/10.1021/jacs.2c01819
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of fragment **27** and the tag-modified compound **28** was achieved using HATU and HOAt in the presence of DIPEA, followed by removal of Fmoc with piperidine and DBU to afford **29** (Figures S24). Under the same coupling reaction conditions, the fragment **30** was effectively coupled with **29** to give the protected glycopeptide **31** with the hydrophobic tag in an overall yield of 60% over three steps as a precipitate. Deprotection of **31** was accomplished by employing the standard conditions as in the preparation of **1** to provide the desired MUC1 glycopeptide **17** bearing a Tn epitope. Subsequently, compound **17** was subjected to successively enzyme-catalyzed reaction cycles using C1GalT1, ST3Gal1, and ST6GalNAc1 to give MUC1 glycopeptides **18**−**20** with a mercapto-functionalized spacer, respectively. Due to the enzyme-catalyzed reaction performed in the aqueous phase and the purification performed by rapid C18 solid-phase extraction, the resulting MUC1 glycopeptides **18**−**20** bearing T, α2,3-sialyl T, and disialylated T epitope could be randomly taken out to be analyzed by liquid chromatography−mass spectrometry (LC-MS) after each enzymatic transformation. As shown in Scheme 2B, each conversion was efficiently carried out.

Convergent Chemoenzymatic Synthesis of a 31-Mer Glycosylated Glucagon-Like Peptide-1. Inspired by the successful assembly of complex MUC1 glycopeptides, a longer 31-mer glycosylated glucagon-like peptide-1 (**21**) bearing an α2,3-sialyl T epitope, which was shown to significantly improve its half-life and pharmacokinetic property,[58] was selected to demonstrate the robustness of this technology. First, we rationally dissected the target peptide sequence into four fragments **32**−**35**, which were efficiently assembled using the standard conditions as mentioned above (Scheme 3 and Figures S32−S35). Next, the resulting glycopeptide **36** was further glycosylated by two sequential enzyme-catalyzed reaction cycles using C1GalT1 and ST3Gal1 to give the desired glycopeptide **21**. All synthesized glycopeptides were characterized by LC-MS (Figures S4−S19, S27−S30, S38, and S39) and further confirmed by HRMS (Table S7).

CONCLUSIONS

Glycosylations of peptides or proteins implicated in many human pathological processes have been identified by advanced analytical technologies and genetic engineering technologies.[2] Previous efforts to prepare glycopeptides have commonly relied on chemical synthesis in the solid phase.[20−23,27−29,59] Despite that these methods have been applied to produce specific types of glycopeptides, limitations still exist including the use of relatively larger excesses of building blocks such as difficultly available complex glycosylated amino acids, the obstacle in monitoring the progress of the reaction, and the cumbersome deprotection process.[26,29] The integration of liquid-phase peptide synthesis with glycosyltransferase-catalyzed oligosaccharide extension provides a potential option for solid-phase chemical synthesis for complex glycopeptide assembly. The liquid-phase chemoenzymatic approach described here greatly accelerates the preparation of various complex glycopeptides with the following merits. First, liquid-phase peptide synthesis breaks the scale limitation and reduces the use of building blocks that are not commercially available or very expensive to purchase in larger quantities. Second, the hydrophobic tag-assisted expeditious centrifugation facilitates the product purification procedure in just a few minutes. Third, the glycan moieties are efficiently installed by robust glycosyltransferases, which can be readily expressed,[60,61] and the product is purified through rapid solid-phase extraction using a C18 cartridge, enabling easy access to various complex glycopeptides. The broad applicability of this method was proved by the streamlined synthesis of 16 complex SARS-CoV-2 *O*-glycopeptides, 4 tumor-associated 20-mer MUC1 glycopeptides, and a 31-mer glycosylated glucagon-like peptide-1. Based on the general applicability and robustness of this streamlined methodology, it is possible to implement complex glycopeptide synthesis in many laboratories, thereby greatly facilitating progress in glycoscience.

The target glycopeptides described here are prepared in a range from 1 to 85 μmol (1−105 mg) scales, which are sufficient to serve as standards for structure studies, investigate their biological functions, and develop glycopeptide-based vaccines and therapeutics. The current approach can be applied to reaction scales of more than 105 mg using larger reaction vessels and carrying out reactions and separations in a parallel way. Although various complex glycopeptides can be readily assembled using this integrated liquid-phase chemoenzymatic approach, peptide aggregation is still worthy of attention. To avoid aggregation during peptide synthesis, strategies including temperature alteration, solvent adjustment, and rational peptide dissection can be applied to further expand the scope of this approach.[62] Furthermore, not every monosaccharide can be randomly introduced into the peptide chain due to the lack of unidentified glycosyltransferases. Advances in genetic engineering technologies and recombinant protein expressions will provide chances to generate more glycosyltransferases, facilitating the diversification and capability of enzyme-catalyzed glycan installment. Additionally, the approach presented in this work has the potential to develop a liquid-phase glycopeptide synthesizer by the combination of artificial intelligence and mechanical manufacturing technologies[33,63] for democratizing synthesis of various complex glycopeptides of biological importance.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.2c01819.

Experimental information, product characterization, and LC-MS and HRMS spectra (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This research was financially supported by the National Natural Science Foundation of China (22077130) and Shanghai Municipal Science and Technology Major Project. We thank Dr. Lin Liu (CCRC, UGA) for proofreading the article.

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https://doi.org/10.1021/jacs.2c01819
J. Am. Chem. Soc. 2022, 144, 9057−9065